58) Show that $p\mid(q\mid r)$ and $(p\mid q)\mid r$ are not equivalent, so that the logical operator $\mid$ is not associative.

Soln: $p\mid(q\mid r) \equiv p\mid(\neg(q\wedge r))$ (By Ex-51)

$\equiv p\mid(\neg q\vee\neg r)$ (By 1st De Morgan's Law)

$\equiv \neg(p\wedge(\neg q\vee\neg r))$ (By Ex-51)

$\equiv \neg p\vee\neg(\neg q\vee\neg r)$ (By 1st De Morgan's Law)

$\equiv \neg p\vee(\neg(\neg q)\wedge\neg(\neg r))$ (By 2nd De Morgan's Law)

$\equiv \neg p\vee(q\wedge r)$ (By Double negation law)

$\equiv (\neg p\vee q)\wedge(\neg p\vee r)$ (1st Distributive Law)

$(p\mid q)\mid r \equiv \neg(p\wedge q)\mid r$ (By Ex-51)

$\equiv (\neg p\vee\neg q)\mid r$ (By 1st De Morgan's Law)

$\equiv \neg((\neg p\vee\neg q)\wedge r)$ (By Ex-51)

$\equiv \neg((\neg p\vee\neg q)\vee\neg r$ (By 1st De Morgan's Law)

$\equiv \neg(\cancel{\neg}\neg p\vee\neg q)\vee\neg r$ (By 1st De Morgan's Law)

$\equiv \neg(\neg(p\wedge q))\vee\neg r$ (By 1st De Morgan's Law)

$\equiv (p\wedge q)\vee\neg r$ (Double-negation Law)

$\equiv \neg r\vee(p\wedge q)$ (1st commutative Law)

$\equiv (\neg r\vee p)\wedge(\neg r\vee q)$ (1st Distributive Law)

~~Let $(\neg p\vee q)\wedge(\neg p\vee r)$ be True~~

p	q	r	$\neg p$	$\neg p\vee q$	$\neg p\vee r$	$(\neg p\vee q)\wedge(\neg p\vee r)$	$\neg r$	$\neg r\vee p$	$\neg r\vee q$
T	T	T	F	T	T	T	F	T	T
T	T	F	F	T	F	F	T	T	T
T	F	T	F	F	T	F	F	T	F
T	F	F	F	F	F	F	T	T	T
F	T	T	T	T	T	T	F	F	T
F	T	F	T	T	T	T	T	T	T
F	F	T	T	T	T	T	F	F	F
F	F	F	T	T	T	T	T	T	T

$\neg(r\vee p)\wedge(\neg r\vee q)$
T
F
T
F

F T

It's easy to see from the truth table that $(\neg p\vee q)\wedge(\neg p\vee r)$ is not logically equivalent to $(\neg r\vee p)\wedge(\neg r\vee q)$. $\therefore p\mid(q\mid r)$ is not logically equivalent to $(p\mid q)\mid r$ $\square$.